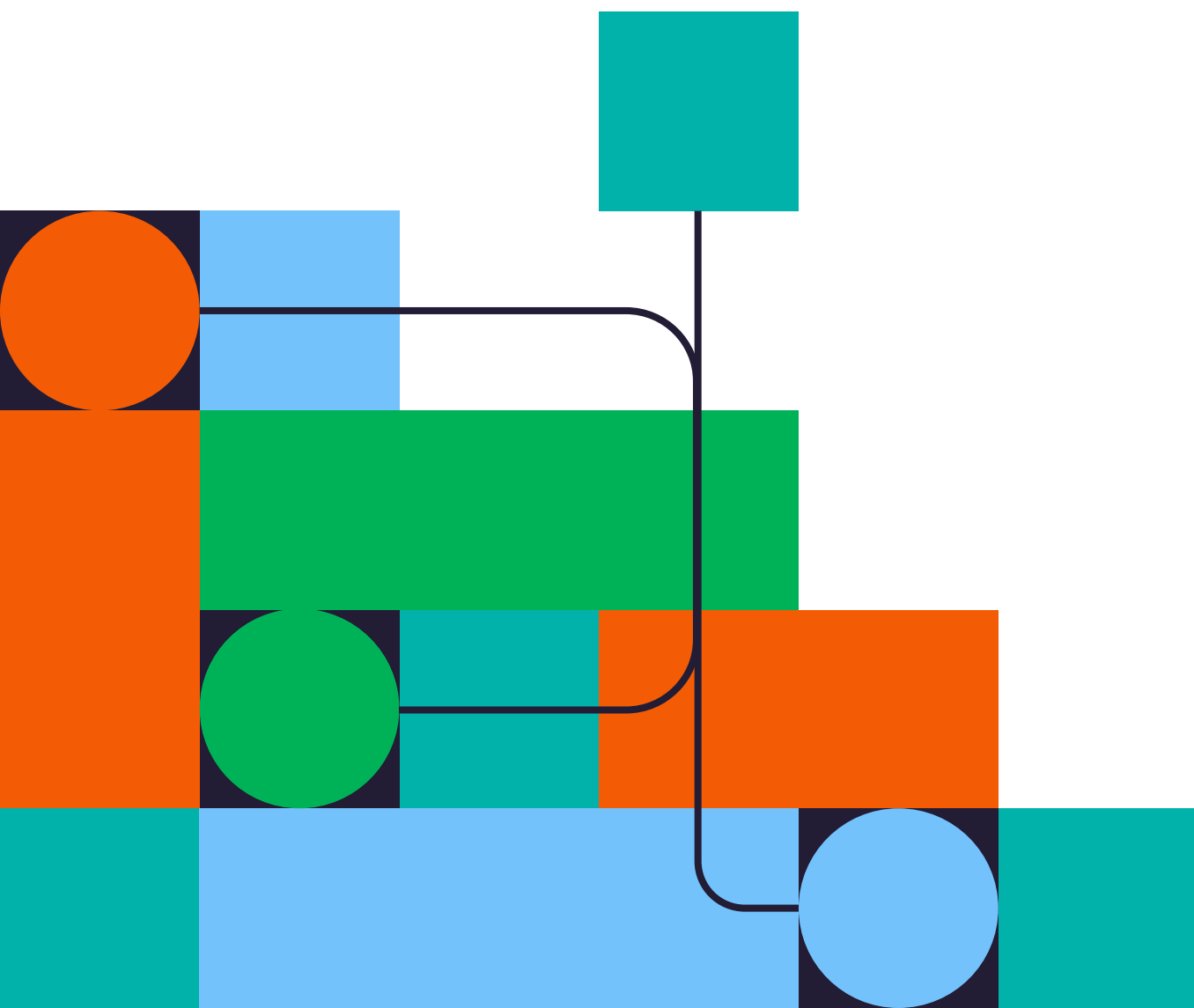


E-MAGAZINE

Generative AI Perspectives From Databricks, Deloitte, & NVIDIA

 databricks | **Deloitte.** |  **NVIDIA.**



Foreword

Generative AI will be transformative, with the potential for embedding human-like intelligence in countless processes and decisions. At the same time, it will complement — not replace — existing analytics and machine learning. It is an area of very active research and development, but has real, immediate applications that enterprises should be working on already. You should thus start now, but expect continuous evolution and innovation, loosely coupling your applications to underlying infrastructure and providers.

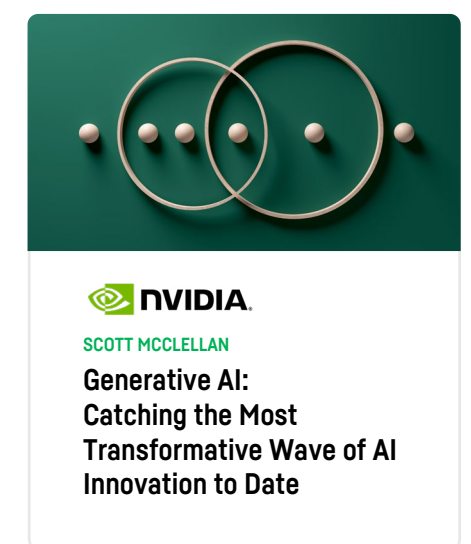
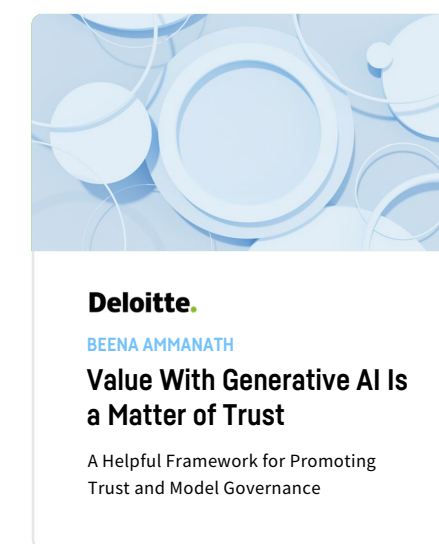
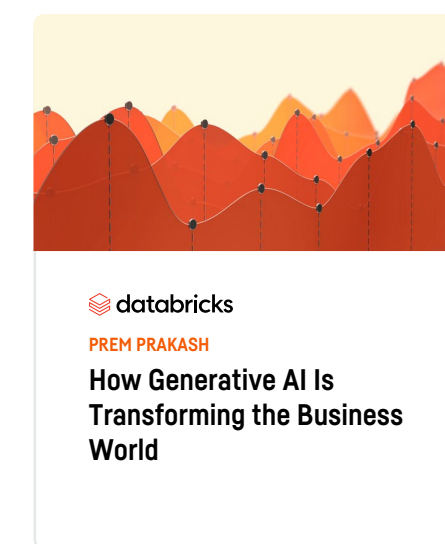
With Dataiku — the platform for Everyday AI — teams can move beyond the lab and build real and safe Generative AI applications at enterprise scale. Dataiku brings enterprise-grade development tools, pre-built use cases, and AI-powered assistants to help everyone do more with Generative AI.

A pivotal piece of that puzzle is our ecosystem, from integrations with leading Generative AI providers to partnerships with leading consulting firms. With our model- and provider-agnostic approach, teams will always be able to leverage the most cutting-edge technologies, enabling maximum agility to meet their business needs.

Just as Dataiku has solidified its position in the data ecosystem as the common backbone for Generative AI applications (notably via the LLM Mesh), Databricks, Deloitte, and NVIDIA underscore the critical importance of that ecosystem and are here as our trusted partners to share their various points of view and subject matter expertise on Generative AI.

- Clément Stenac

Co-Founder and CTO, Dataiku



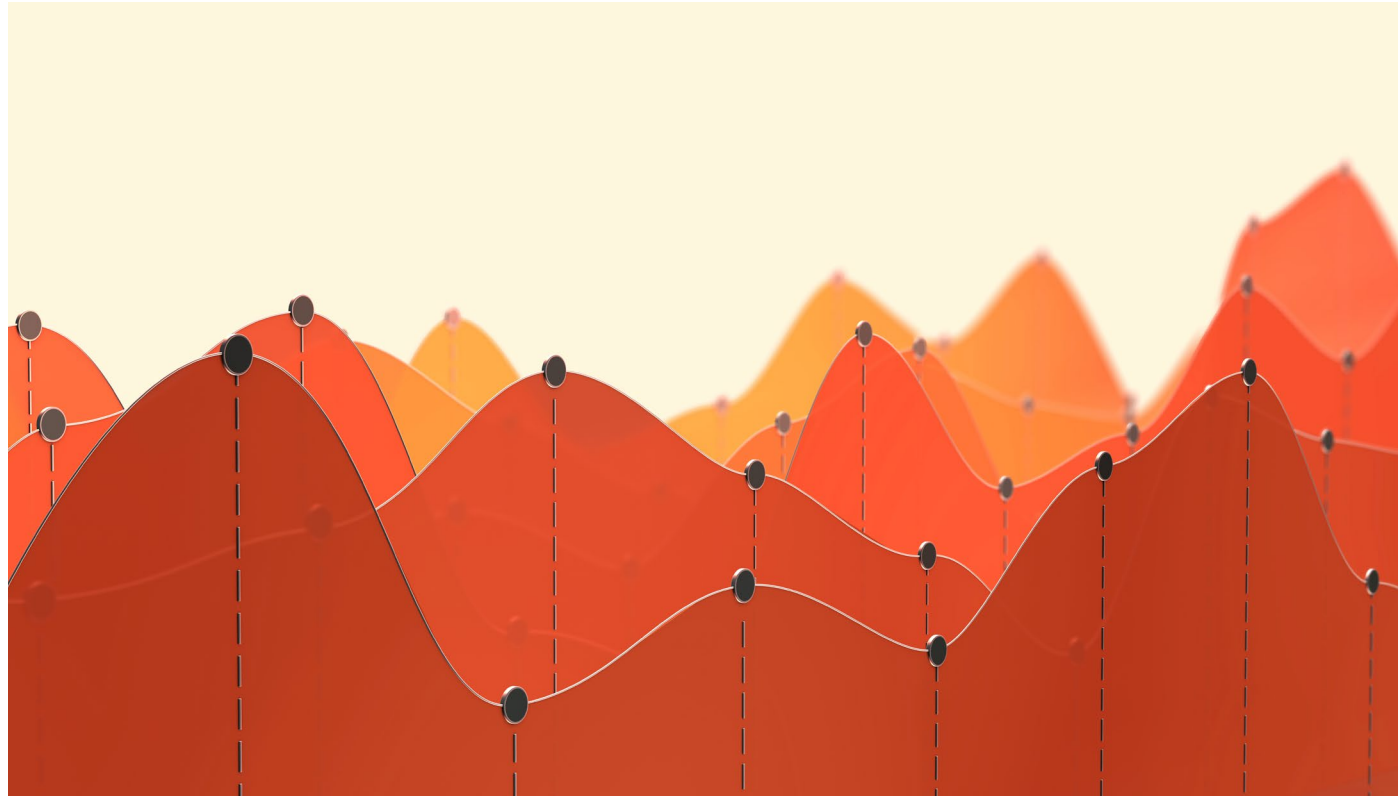


How Generative AI Is Transforming the Business World



Prem Prakash Bio

Prem Prakash is the Head of Product Marketing for AI, Developer Experience and Learning at Databricks. He previously led Product Marketing for Azure AI services at Microsoft.



A monumental shift is underway in today's dynamic business environment, primarily driven by Generative AI. As we delve deeper into this digital transformation era, Generative AI stands

out as a game changer, steering businesses toward uncharted territories and innovative possibilities.

Let's unpack the multifaceted impact of Generative AI on modern business.

The Rise of Generative AI in Business Operations

Gone are the days when AI was a buzzword that echoed in tech forums and conferences. Today, it is a cornerstone of innovation, transforming day-to-day business operations. Through enhanced and deeply engaging customer experiences, automation of routine tasks, and accelerated decision making, Generative AI has firmly established its footprint across sectors.

Every sphere, whether it's healthcare, finance, retail, or real estate, is feeling the transformative

impact of Generative AI. It can reshape conventional business models, from refining supply chain logistics to offering real-time analytics and creating unparalleled customer experiences.

Due to this unbounded potential, industry analysts estimate the benefits of Generative AI to be in the trillions of dollars. **McKinsey Global Institute estimates that Generative AI will add between \$2.6 and \$4.4 trillion in annual value to the global economy.**

“

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An intriguing revelation from a joint study from MIT and Databricks shows that 88% of business leaders are tilting their investments toward Generative AI.

The Investment Surge in Generative AI

While several IT sectors grapple with budget constraints, the appeal of AI investments is hard to ignore. Despite financial hurdles, there's a growing inclination toward Generative AI, underscoring its recognized enduring worth.

An intriguing revelation from a [joint study from MIT and Databricks](#) shows that 88% of business leaders are tilting their investments toward Generative AI¹. This isn't a fleeting trend; it reflects AI's vast transformative potential.

Furthermore, these investments aren't just whimsical gambles. Business executives anticipate impressive dividends, spanning amplified operational efficiencies, substantial cost curtailments, and groundbreaking innovations. This confidence underscores the deep-seated faith in Generative AI's capacity to spearhead revolutionary changes across industries.

¹ MIT and Databricks. (2023). (rep.). *The Great Acceleration – Perspectives on Generative AI*. Retrieved from <https://www.databricks.com/resources/ebook/mit-cio-generative-ai-report>.



ELEVATING CUSTOMER INTERACTIONS

The recent research spearheaded by MIT and Databricks sheds light on the future of Generative AI, emphasizing its pivotal role in customer experience and personalization for the upcoming years. Here's how Generative AI stands to revamp the customer engagement paradigm:

- **Employee Productivity:** Generative AI can automate manual and repetitive tasks, be a personal assistant (to organize calendar or manage emails), and improve the pace of work by suggesting code to write for developers or content to write for marketers.
- **Enhanced Human Interactions:** Generative AI can craft text that mirrors human conversations, adapting to various languages, conversational nuances, and emotional undertones. Further, its integration into search engines ensures swift and accurate answers, thus streamlining user queries.
- **Leveraging Customer Data:** By imbibing customer-specific data into models, organizations can proactively discern, foresee, and cater to the unique requirements of individual clients, patients, or stakeholders.
- **Tailored Customer Engagements:** Whether it's policyholders nearing renewal deadlines, citizens settling taxes, or patients up for health check-ups, Generative AI offers automated solutions. These span from policy renewals and timely appointments to hassle-free product returns.
- **Automated Document Management:** The technology can autonomously draft or refresh critical documents, such as medical reports or contracts, without human oversight.
- **Iterative Customer Engagement:** The models constantly assimilate feedback, ensuring that subsequent customer interactions are continually refined based on past experiences.

STRENGTHENING CYBERSECURITY

In today's interconnected digital realm, safeguarding sensitive organizational data is critical. Generative AI emerges as a linchpin in streamlining and bolstering cybersecurity measures.

- **Automated Protection Mechanisms:** By automating tasks like threat detection and real-time responses, Generative AI mitigates risks associated with human errors — a significant cause of data breaches.
- **Empowering Cybersecurity Teams:** A consolidated AI platform democratizes the technology, which means even analysts with foundational skills can glean rapid insights, optimizing costs and freeing seasoned experts to innovate further.
- **Proactive Code Security:** Generative AI's capabilities include scanning infrastructure codes, pinpointing vulnerabilities, and initiating automated remediations. This not only maintains code quality but also preempts potential security threats.

While enterprises often commence their Generative AI voyage eyeing profit augmentation, ensuring robust cybersecurity is paramount. Neglect here could lead to spiraling costs tied to cyber incidents, tarnished brand image, customer attrition, and regulatory penalties.



MODERNIZING SUPPLY CHAIN OPERATIONS

Generative AI's capabilities extend to optimizing supply chain mechanisms. By assimilating and synchronizing data throughout the supply chain, it bridges the gap between upstream anticipations and downstream deliveries.

- **Precision Forecasting:** Generative AI mines historical data and market tendencies, aligning them with inventory stats, supplier delays, and output to determine intricate patterns and unveil opportunities often missed by traditional methods.
- **Strategic Supplier Engagement:** The technology aids in selecting optimum suppliers based on performance metrics, quality benchmarks, and cost-effectiveness. Furthermore, it equips enterprises with insights to manage these supplier relations effectively.
- **Predictive Maintenance Enhancement:** Generative AI's feedback loop adjusts to equipment updates, environmental variations, and usage patterns. This ensures timely equipment maintenance and proactive response mechanisms.



Industry leaders unanimously concur that Generative AI can yield dual benefits: substantial cost reductions and novel revenue streams. This is achievable by enhancing operational efficiency, expediting insights derivation, broadening visibility scopes, and underpinning decisions with data-driven methodologies.

Generative AI's Potential

Generative AI, once an aspirational concept, has now matured into a potent tool with tangible applications. Many open source and easily customizable models are now within the reach of businesses, marking a pivotal moment in AI's evolutionary journey. Notably, functionalities that seemed distant a mere year ago are now seamlessly executed by business users, even those without deep technical know-how.

In the aforementioned collaborative research initiative with MIT, Databricks discerned a trend of burgeoning optimism. **A staggering 80% of surveyed executives are confident that AI will amplify operational efficiency by a minimum of 25% within the next two years.** Moreover, a substantial one-third project this uptick to reach or exceed 50%. Such ambitious projections underline the enthusiasm surrounding Generative AI and its transformative potential across corporate landscapes.

Industry leaders unanimously concur that Generative AI can yield dual benefits: substantial cost reductions and novel revenue streams. This is achievable by enhancing operational efficiency, expediting insights derivation, broadening visibility scopes, and underpinning decisions with data-driven methodologies.

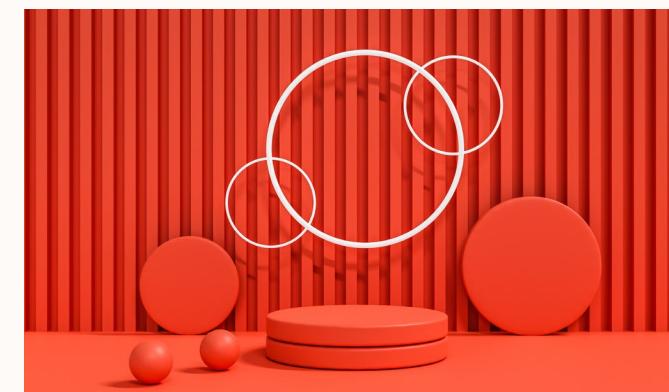
Given the impressive ROI trajectories associated with Generative AI, it's hardly surprising that 88% of executives are investing in it. Furthermore, over a quarter of those executives are already in the implementation phase.

However, beneath this buoyant outlook lies a consensus: For Generative AI to deliver consistently at an expansive scale and ensure a viable Total Cost of Ownership, certain foundational pillars are indispensable:

#1

Comprehensive Data Access:

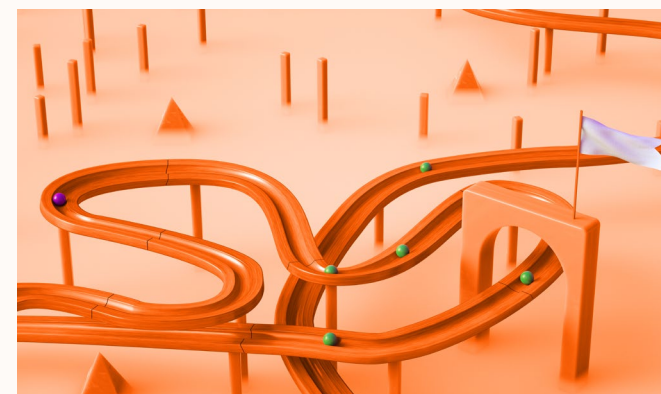
The essence of Generative AI is its ability to tap into diverse data – regardless of its nature, origin, or real-time status.



#2

Robust Data Governance:

Central to any successful AI blueprint is governance. The strength and relevance of AI models hinge upon the integrity, accuracy, and timeliness of the data they draw from. Enterprises must maintain resilient data conduits and update and expand data sources as needed. The AI framework.



#3

Effective Data Sharing:

Realizing the transformative potential of Generative AI mandates capitalizing on the collective strength of an organization's entire ecosystem. By integrating essential third-party data channels, like the Databricks Marketplace, companies can enrich their models. This facilitates swifter, more insightful decision-making, enabling businesses to transition from a passive stance to a proactive one.





Databricks AI: A Data-Centric Approach to Generative AI

For Generative AI to yield impactful outcomes, AI models must grasp the specialized terminologies and distinct data sources characterizing each industry. This underscores the necessity for industries — and even individual entities within them — to approach Generative AI in a manner that's uniquely tailored, considering how their specific data sets can influence the creation of Generative AI tools.

What stands out is the need for businesses to utilize their distinctive data to carve out Generative AI strategies that set them apart in the marketplace.

Enter Databricks AI. With its specialized emphasis on data, Databricks AI equips users with the tools to craft and launch AI models swiftly, efficiently, and with a robust governance framework. This allows businesses to use their proprietary data as a starting point, swiftly constructing and introducing generative AI tools tailored to their sector and niche.

Be it the aspiration to create Q&A chatbots leveraging open source LLMs combined with Retrieval Augmented Generation (RAG) or the ambition to devise bespoke models by fine-tuning an existing pre-trained LLM with proprietary data, Databricks AI is equipped for it all.

With Databricks, you also get LLMOps which is management of all aspects of the lifecycle—from data ingestion, model building, tuning, and productionization — from a single platform. A notable feature is its innate governance and monitoring capabilities, ensuring the creation of models within a secure environment where data privacy, permissions, and compliance are paramount.

With Databricks MosaicML, businesses have the advantage of constructing expansive AI models on their proprietary data within their private domains, ensuring swiftness, security, and cost efficiency. MosaicML is the most cost-effective way to fine-tune or pretrain models, up to 7x lower costs than pre-training other Generative AI models.

Unlocking Success

Generative AI is not merely a technological trend; it's a revolution. It's redefining the contours of industries, making operations more innovative, efficient, and customer centric. As we look ahead, businesses that understand, adopt, and innovate with generative AI will not just thrive but lead.

The future is not just automated; it's generative, promising, and tailored to unique needs, ensuring businesses remain relevant, competitive, and successful.

Value With Generative AI Is a Matter of Trust

A Helpful Framework for Promoting
Trust and Model Governance



Beena Ammanath Bio

Beena leads Trustworthy AI & Technology Trust Ethics at Deloitte. She is the author of “Trustworthy AI,” a book that can help businesses navigate trust and ethics in AI. Beena has extensive global experience in AI and digital transformation, spanning across e-commerce, finance, marketing, telecom, retail, software products, services, and industrial domains with companies across a variety of industries.

Beena is also the Founder of the nonprofit, Humans for AI, an organization dedicated to increasing diversity in AI. Beena also serves on the Board of AnitaB.org and the Advisory Board at Cal Poly College of Engineering. Prior to her joining Deloitte, she was a Board Member and Advisor to several technology startups. Beena thrives on envisioning and architecting how data, artificial intelligence, and technology in general, can make our world a better, easier place to live for all humans.



The world has entered a new era with AI.

Beginning in earnest in 2022, the release of several consumer-facing, headline-grabbing Generative AI applications captured public attention and stimulated interest among organizations and across industries. Innovation has come fast, and today, novel Generative AI applications are emerging in areas such as hiring and workforce management, medical diagnosis and treatment recommendations, supply chain efficiency and constraint management in manufacturing, and internal controls for financial reporting.

The potential with Generative AI is grand in part because of how this type of AI functions. Whereas deep learning, for example, is valuable for discovering patterns and making predictions, Generative AI is characterized by its capacity to create data that appears as coherent as its training set. As the marketplace of Generative AI solutions matures, models that can create text, images, code, video, and much more will serve as an enabler of work and creativity.

Of course, new technologies often conjure new questions around risk and trust. Even as organizations are contemplating how to leverage Generative AI for efficiency gains, tailored offerings, product improvements, new experiences, and business transformation, attention should also be given to the governance and risk management that makes these use cases trustworthy and valuable.

To pursue confidence and trust in Generative AI, there are a variety of considerations and strategic actions that can help govern the model lifecycle and balance commercialization, regulation, ethics, co-creation, risk management, and ultimately, trust. The most impactful approaches are found in the familiar areas of people, processes, and technologies.

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Preparing the Workforce for the Generative AI Era

The technical function of Generative AI models may only be deeply understood by the organization's data and IT professionals, but governing AI is a whole-of-enterprise activity. The onus for identifying and managing risk is shared throughout the organization, including with implementation leads, development leads, data governance managers, peer reviewers, security experts, and line of business users.

What is more, in a time when workforces are evolving, stakeholders can also include vendors, partners, and contingent workers. For these professionals to take part in Generative AI governance, they require a clear understanding of their roles and responsibilities in AI development, deployment, and outcomes.

Stakeholders also require AI literacy as it relates to Generative AI applications. They need a working understanding of the AI model and its use cases, its drawbacks and limitations, as well as the risks it may create. To promote Generative AI literacy, business leaders can leverage existing workforce training and knowledge transfer sessions between technical and non-technical stakeholders.

The organization may also offer or promote online education, industry conferences, or seminars, all to nurture a culture of continuous learning. This is particularly important with Generative AI, a technology that is changing and improving at a rapid pace. Taking part in risk management and governance requires stakeholders to keep up with developments in Generative AI.

As well as AI literacy, stakeholders may also need new skills specific to their practice area:

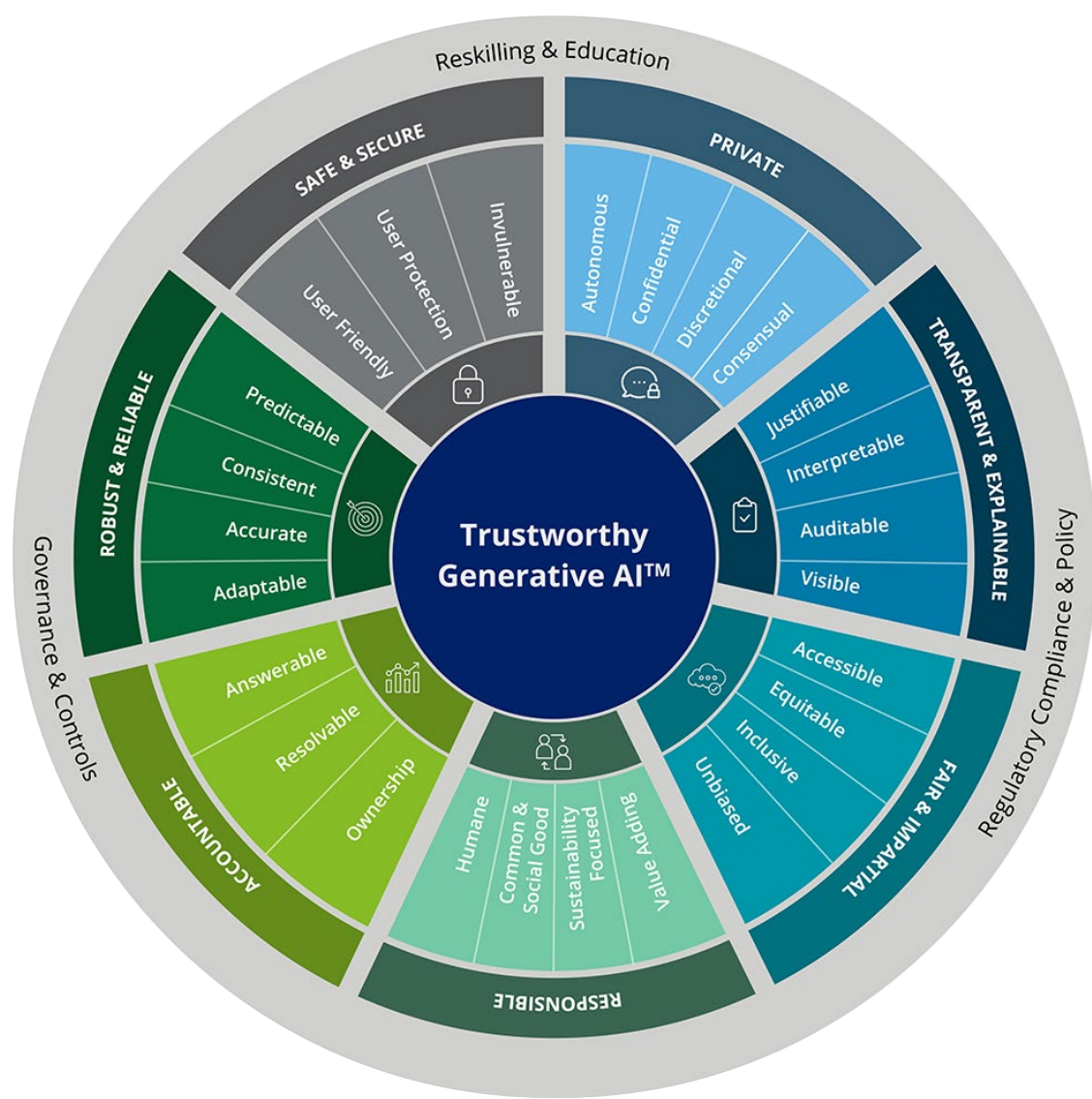
- Compliance and regulatory leads need familiarity with the technology so as to understand the potential impact from AI laws and regulations emerging in regions where the business operates.
- Marketing and communications professionals need an understanding of how to leverage Generative AI capabilities given questions around intellectual property and copyrights.
- Human relations leaders need to see how Generative AI applications can improve workforce programs, employee support, and third-party engagement.
- Business domain experts need to collaborate with computer and data scientists to translate business objectives into value-driving AI use cases.
- The C-suite and the organization's board need to understand the vision for how Generative AI enhances the business and the risks it may create along the way.

Assessing New Risks to Inform New Processes

Generative AI will likely be an integral part of the future of work, with widespread adoption across industries and lines of business. This will amplify human-machine collaboration in the Age of With™, as Generative AI enables tools that supplement and enhance the human workforce. This will lead to changes in how work is accomplished, which has implications for business processes and workflows that support Generative AI risk management and governance.

The first step is to understand the risk domains and where gaps exist by conducting a risk assessment and analysis. No two AI applications are precisely the same, fueled as they are by proprietary enterprise data. In conducting risk assessments, a helpful tool is Deloitte’s Trustworthy AI™ framework, which helps the enterprise identify elements of trust that may apply to a Generative AI deployment.

A trustworthy use case is assessed for its capacity to be fair and impartial, robust and reliable, transparent and explainable, safe and secure, accountable and responsible, and respectful of privacy. Identifying which of these risk domains is relevant to a given use case serves as the basis for designing risk mitigation controls, responses, and other AI Governance processes. It also informs key governance questions, including:



- **DATA GOVERNANCE AND PROCESSING:** How does the organization obtain, store, transfer, and curate the data used for Generative AI? Is third-party data free of bias, and how is that determined? Has personally identifiable information been appropriately obscured so that it does not “leak” through the deployment?
- **SECURITY:** How does the organization secure data used by AI models, as well as the models themselves? What kinds of security vulnerabilities exist, and what are the tactics a bad actor may employ to compromise the model (e.g., prompt-spoofing)? At which points in the Generative AI lifecycle is safety and security addressed, and how is it assessed and maintained over time?
- **ORGANIZATIONAL GOVERNANCE:** How does the organization define testing objectives? How does it monitor and evaluate results from controls and processes? How are stakeholders empowered to provide input, feedback, and concerns throughout the AI lifecycle? Are there additional roles that could be created to support stakeholder participation and input, such as an AI ethics board or even a Chief AI Ethics Officer?
- **MODEL DEVELOPMENT AND EVALUATION:** How does the organization validate AI model performance? Who is involved in that endeavor, and what tools and technology platforms can help?



By deeply understanding how a model performs, an organization can enhance other aspects of Generative AI Governance, including necessary workforce upskilling, stakeholder engagement, and process transformation.

Leveraging Technology for Model Governance

Today's Generative AI models often come with a disclaimer that the outputs may be inaccurate. The model itself would be forgiven as Generative AI creates data that looks like real data. It has no capacity to understand whether the generated data is valid. That task falls on the human stakeholders.

The issue of reliability points to a larger challenge in AI Governance, which is how to understand, evaluate, and measure the performance of a tool whose technical function may not always be understandable. The “black box” problem with other types of AI is perhaps even more perplexing with Generative AI models. Large language models, for example, can have billions of parameters. The scale defies human comprehension.

To help ensure accountability, organizations may leverage technology solutions that help evaluate and track model performance. This allows the enterprise to evaluate whether an AI tool performs as intended and aligns with the relevant dimensions of trust.

Importantly, the product of technology-enabled assessments is not just improvements to the model itself. By deeply understanding how a model performs, an organization can enhance other aspects of Generative AI Governance, including necessary workforce upskilling, stakeholder engagement, and process transformation.

Looking Ahead to the Future With Generative AI

These approaches for orienting people, processes, and technologies can help organizations operationalize trustworthy Generative AI.

However, in an increasingly crowded marketplace of solutions, organizations are challenged to not only implement governance for these powerful new tools but also keep up with advances and even anticipate capabilities and use cases just over the horizon. A professional services collaborator with deep experience in trustworthy AI development and application can offer critical support as the enterprise establishes Generative AI Governance.

Deloitte helps clients take a three-pronged approach to Generative AI Governance:

- Evaluating roles and responsibilities, supporting change management, and conducting workforce training.
- Offering services around AI strategy and governance, risk management, and AI assessments.
- Using tools to monitor AI function and performance, such as Dataiku's platform for assessing, managing, and documenting each step of model development and operation.

Across the spectrum of professional services — from strategy consulting to advisory support to audit and assurance — Deloitte can offer the approach and offerings that help you take the right steps, make sound decisions, and implement the processes and programs that help you maximize value and trustworthiness with Generative AI.

Sources:

- [Responsible Use of Generative AI](#)
- [Trustworthy AI Governance in Practice](#)
- [Services: Trustworthy AI™](#)
- [Report: Trustworthy AI in Practice](#)
- [Article: Proactive Risk Management in Generative AI](#)
- [Article: The Implications of Generative AI for Businesses](#)
- [Document: Generative AI Is All the Rage](#)
- [Article: Benefits and Limitations of Generative AI](#)

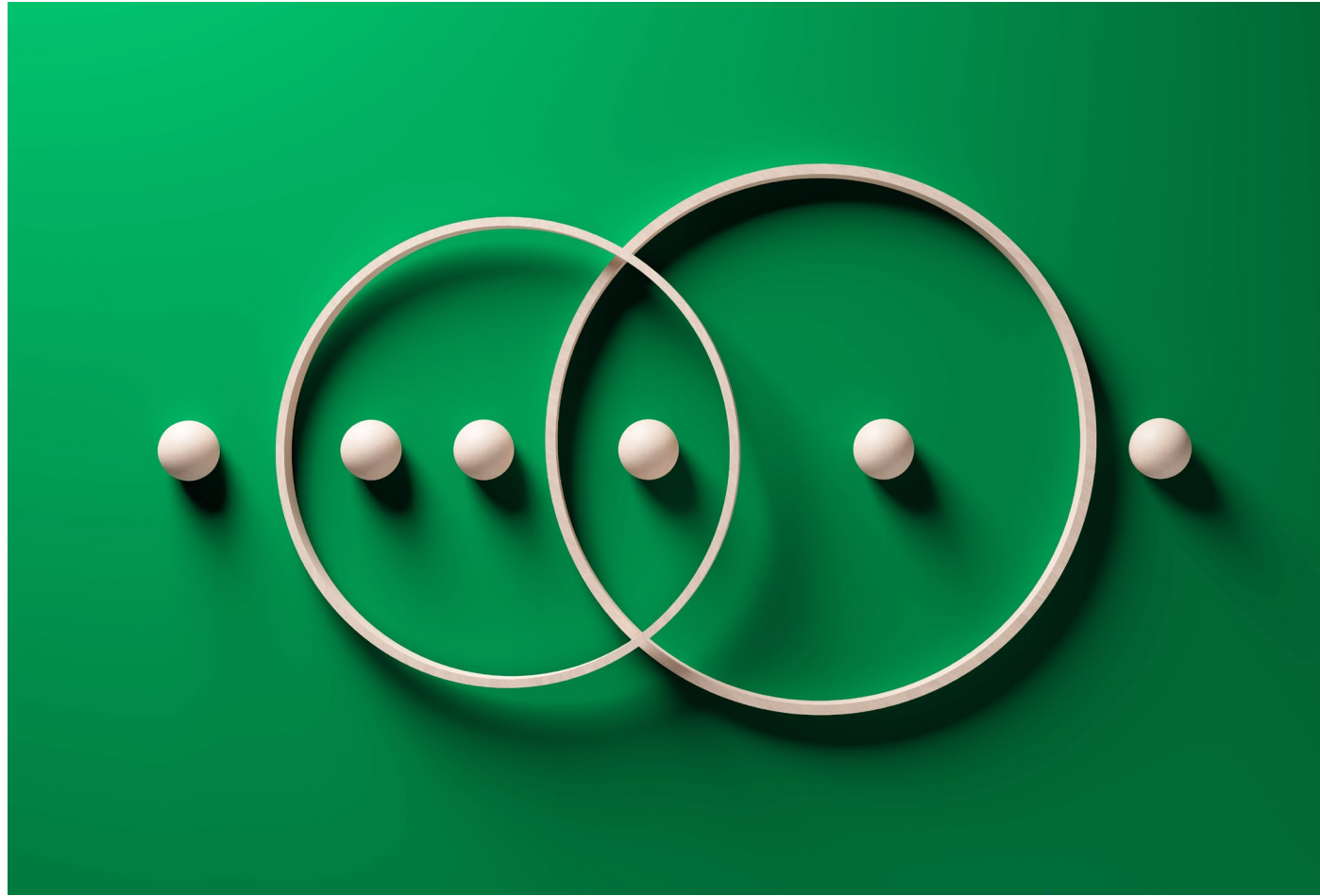


Generative AI: Catching the Most Transformative Wave of AI Innovation to Date



Scott McClellan Bio

Scott McClellan is a senior director of product management at NVIDIA, focused on data science and other AI workflows. Before joining NVIDIA, Scott was the chief technology officer of PRGX Inc. He has been chief technologist and led engineering and product development at companies including RedHat and HP, where he guided strategies across HPC, cloud, big data and AI solutions. Scott holds a Bachelor of Science from the University of Iowa.



Today, nearly every executive or decision maker responsible for IT or application technology strategy has seen an avalanche of excitement about Generative AI or Large Language Models (LLMs).

Generative AI and LLMs represent two of the latest waves of innovation in AI research. This feature gives an overview of both technologies, highlighting some key considerations for safely surfing the Generative AI/LLM technology wave.

Generative AI

[Generative AI](#) refers to a class of AI capable of generating **net new content** based on a variety of inputs. The ability to create an artifact that does not exist in the training data explicitly is the fundamental distinction between Generative AI and more traditional AI.

FOUNDATION MODELS

A Generative AI system “*learns*” underlying characteristics from training data and will use that “*knowledge*” to generate a model based on those characteristics — this model is called a “*foundation model*.” The amount of data required to train a foundation model varies but is typically large, often involving millions or even billions of examples, to achieve high performance and generalization across diverse tasks.

Foundation models have broad, general capabilities and can be customized to perform specialized tasks. The potential need for customization and the methods for customizing models are discussed in more detail later in this feature.

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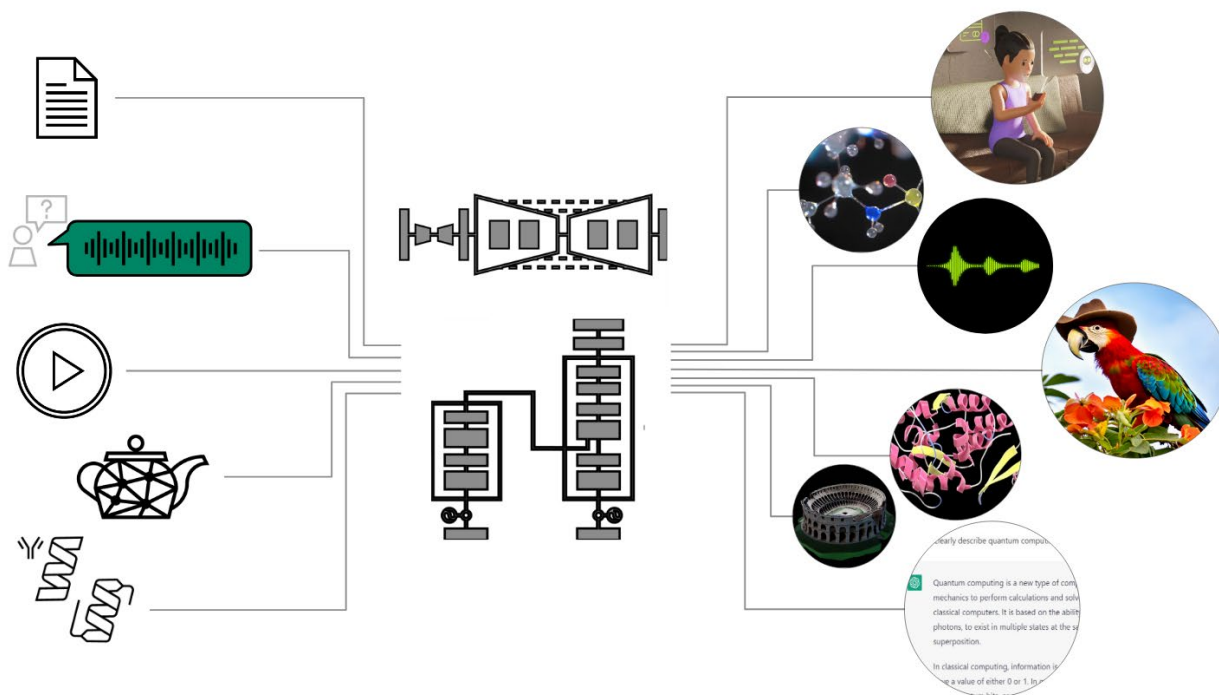


Figure 1: Generative AI

TRANSFORMER ARCHITECTURES

Foundation models are based on “*transformer architectures*,” a type of neural network designed to take an input sequence and use it to generate an output sequence one element at a time.

For example, a transformer could be used to translate a sentence in English into a sentence in French. The process of translation could take the sequence of words in English as input, and the transformer would “*predict*” a corresponding sequence of words in French.

The key innovation in a transformer architecture is the self-attention mechanism, which allows the model to weigh the importance of different parts of the input data when making predictions, improving the quality/fidelity of predictions at a given computational budget.

MULTIPLE MODALITIES

Inputs and outputs to these models can include text, videos, sounds, images, animation, 3D models, or other types of data.

Generative AI can be multi-modal, meaning a given AI pipeline can be designed to understand input data in multiple or mixed formats and can generate output in multiple or mixed formats.

Large Language Models

LLMs are a form of Generative AI capable of generating coherent and relevant text based on the input they receive. These models can be used for various natural language generation tasks, such as text summarization, language translation, and even creative writing.

In the case of an LLM, the foundation model is trained on massive amounts of diverse text data. The use of self-supervised training has drastically expanded the scale and scope of data LLMs are trained on, effectively allowing foundation models to “*learn*” from the vast and diverse information available on the internet. Even with the most advanced techniques for parallelizing the training, and taking advantage of GPUs to greatly accelerate the training process, training a foundation model on “*internet scale*” data takes a long time (days to weeks) and costs millions of dollars.

Using an LLM in an application involves taking a sequence of data called a *prompt*, feeding that prompt to the LLM (in a process called *inference*), and getting back a generated sequence of text which is the model’s response to that prompt.

The LLM generates the response by predicting a new sequence based on the prompt it was given as input. The response is *new content* generated by the LLM. This process differs from search, where a search engine finds existing text that matches instead of generating new text as a response.

Through this process, LLMs are able to capture the underlying patterns and semantic relationships, gain a contextual understanding of language, and use that to generate very human-like responses.

“**Even with the most advanced techniques for parallelizing the training, and taking advantage of GPUs to greatly accelerate the training process, training a foundation model on 'internet scale' data takes a long time (days to weeks) and costs millions of dollars.**”

The ChatGPT Effect

The watershed moment for many major technology trends can be traced to the availability of a “killer application” which showcases the power and value of that technology. OpenAI’s ChatGPT application made it possible for anyone to ask ChatGPT their favorite “stumper” question.

HUMAN-LIKE QUALITIES

Skeptical users would not expect a computer program to demonstrate human qualities such as reasoning, intuition, empathy, humor, creativity, etc. For example, asking ChatGPT to write a poem or a short story.

Note how ChatGPT was able to take a very simple prompt (which included a typo — “poen” not “poem”) and was able to generate a poem about Maui that stayed within the requested word count.

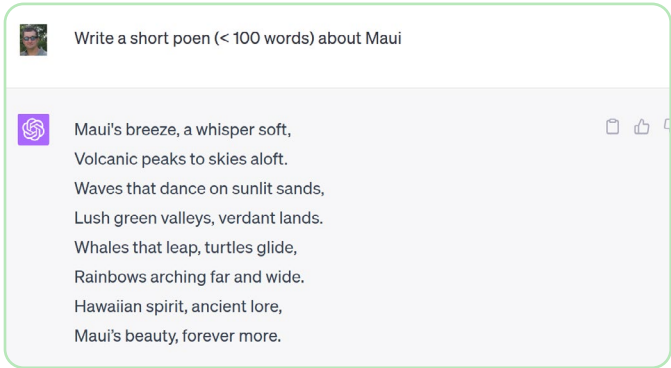


Figure 2: Example poem generated by ChatGPT

TECHNICAL SKILL, DOMAIN EXPERTISE

Similarly, users could choose a topic that required technical or domain expertise — picking a question they were confident ChatGPT could not answer. Below is an example where ChatGPT is asked to write code in the programming language Rust.

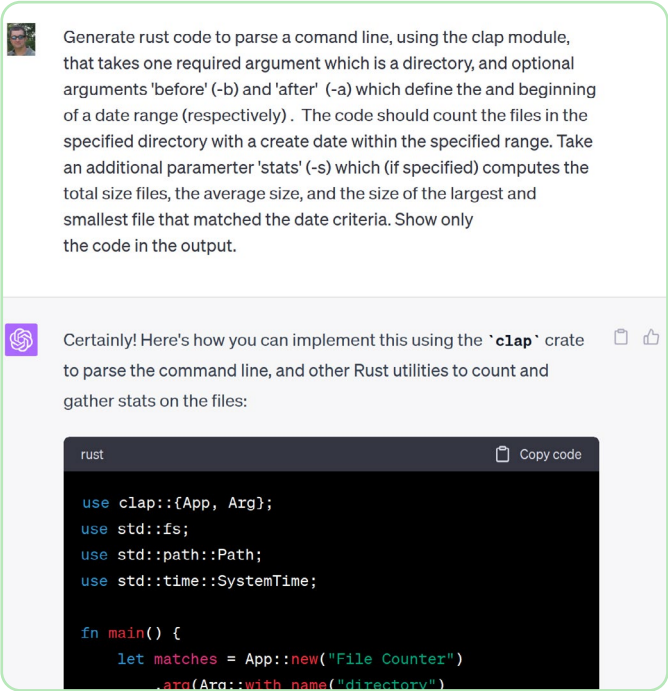


Figure 3: ChatGPT code generation example

The ChatGPT output above was truncated, but the prompt generates working code, as well as a description of how to install and use the code.

This example illustrates a simple form of prompt engineering, using a detailed prompt, delivered to ChatGPT to generate functionally complete source code. The prompt is not perfect but ChatGPT clearly understood what the user wanted. The user could also ask ChatGPT to modify the code, or ask for an explanation of the code, showcasing how an application built on LLMs can be very conversational and natural to use.





How quickly and easily you can take advantage of LLM will depend on how complicated your use case is and the degree to which you need to customize.

Use Cases

The consensus is that Generative AI has the potential to completely transform many enterprises, creating competitive differentiation, improving existing business, and in some cases creating whole new business opportunities.

Below are a few examples of enterprise use cases for AI by industry.

FINANCE Fraud Detection Personalized Banking Investment Insight	HEALTHCARE Molecule Simulation Drug Discovery Clinical Trial Data Analysis
MEDIA & ENTERTAINMENT Character Development Video Editing & Image Creation Style Augmentation	MANUFACTURING Factory Simulation Product Design Predictive Maintenance
RETAIL Personalized Shopping Automated Catalog Descriptions Automatic Price Optimization	MANUFACTURING AI Virtual Assistants Network Performance Tuning Remote Support Capabilities
FEDERAL Document Summarization Audit Compliance AI Virtual Assistants	ENERGY Knowledge Base Q&A Predictive Maintenance Customer Service

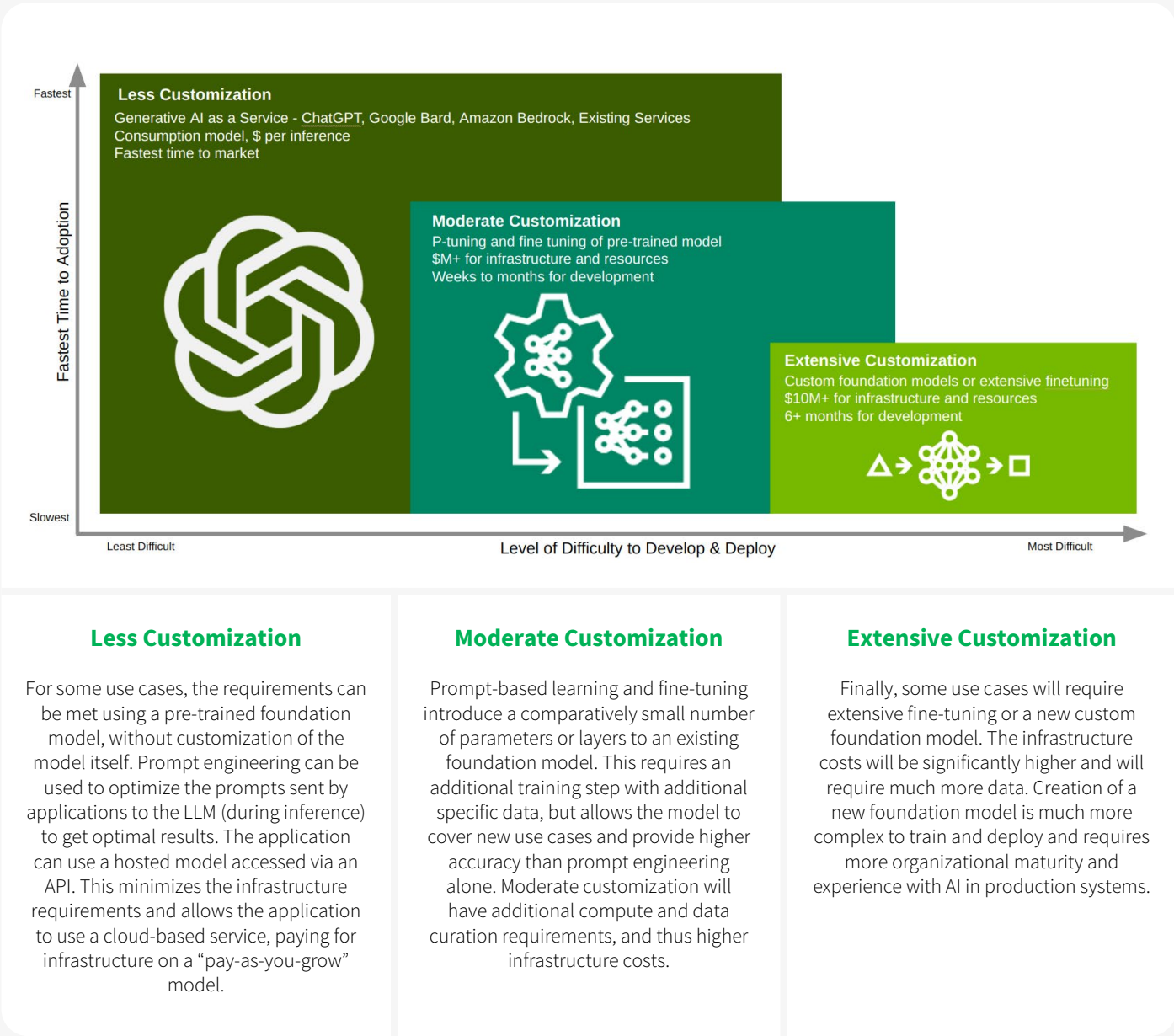
Figure 4: Generative AI Use Cases by Industry

Adoption Considerations

When using any form of Generative AI for a given enterprise use case, it will likely be necessary to do some degree of customization. How quickly and easily you can take advantage of LLMs will depend on how complicated your use case is and the degree to which you need to customize.

CUSTOMIZATION

Here we will characterize use of Generative AI by putting it into one of three buckets based on degree of customization.



AUGMENTING WITH ADDITIONAL DATA

There will be cases where you will want to use additional data to get the desired results.

The simple intuition here is that an LLM can't generate quality output for domains or topics not covered in the data used to train the foundation model. This is particularly true for domains where new data brings new context on a regular basis. Augmenting with additional data adds complexity, but also creates the opportunity for an enterprise to use their own data and domain knowledge as an asset — to provide differentiated value in their application.

For example: If you were building an application to provide advice on building maintenance, and you had thousands of technical documents, maintenance logs, etc., you could fine-tune a pre-trained model with that proprietary data, adding to the knowledge the foundation model already has.

Document Curation

As outlined in the above, there are use cases where you will need to augment an existing pre-trained model, or potentially, create your own foundation model.

Injecting additional data into the workflow, introduces several topics related to curating that data.

DATA VARIETY:

Data from multiple sources, in a multitude of formats.

SECURITY CHALLENGES:

Ensuring valuable proprietary data is not exposed.

DATA PRIVACY:

Ethical and regulatory issues related to private or confidential data.

DATA GOVERNANCE:

The additional data may have different rules that govern how it can and cannot be used.

SCALE CHALLENGES:

Associated with manipulating and storing data for fine-tuning or building your own foundation model.

DATA VERSIONING AND EXPERIMENT MANAGEMENT:

Necessary to support the iterative model development/deployment process.

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Augmenting with additional data adds complexity, but also creates the opportunity for an enterprise to use their own data and domain knowledge as an asset — to provide differentiated value in their application.



NVIDIA AI Solutions

NVIDIA has a complete catalog of software for accelerating all aspects of AI — everything from data processing to traditional machine learning and deep learning training and inference — and a multitude of frameworks to support different general and industry specific use cases. Collectively, this is referred to as [NVIDIA AI software](#).

[NVIDIA NeMo Framework](#) is an end-to-end, cloud-native framework to build, customize, and deploy Generative AI models anywhere. It includes training and inferencing frameworks, guardrail toolkits, data curation tools, and pre-trained models, offering enterprises an easy, cost-effective, and fast way to adopt Generative AI.

NVIDIA NeMo Framework is included with [NVIDIA AI Enterprise](#), the enterprise-grade software that powers the NVIDIA AI platform and accelerates enterprises to the leading-edge AI with security, stability, manageability, and support. NVIDIA AI Enterprise offers extensive frameworks, pretrained models, and development tools to streamline development and deployment of production AI, enabling organizations to harness the game-changing insights of AI instead of maintaining a high-performance software platform.

Click [here](#) to get the latest information from NVIDIA on Generative AI.

Summary

Generative AI represents the latest and most promising breakthrough in AI. In a very real sense, Generative AI is the crowning achievement of decades of research and development in deep learning, building on many of preceding breakthroughs in AI.

LLMs are a specific type of Generative AI, capable of generating human quality responses to textual prompts — fundamentally changing the way humans and computers interact.

Given the vast potential, *all enterprises need to be exploring how to take advantage of Generative AI and LLMs.*

Adoption and deployment of these technologies requires careful consideration, notably with respect to model customization, and operational challenges like data curation.

Solutions like **NVIDIA's NeMo Framework** will help enterprises with these key challenges, and **NVIDIA AI Enterprise** will provide enterprises with the stability, security, and support they need to use these AI in critical production applications.



Dataiku is the platform for Everyday AI, enabling data experts and domain experts to work together to build data into their daily operations, from advanced analytics to Generative AI. Together, they design, develop and deploy new AI capabilities, at all scales and in all industries.



